

A Counterexample to Extending One-Point Asymptotic Beta Positivity Beyond One

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June 26, 2026

Source Question

The source paper is F. P. Baudier and G. Lancien, *An asymptotic analog of a local-to-global phenomenon for uniformly convex renormings*, arXiv:2401.17465. Theorem 1.2 states that if a Banach space $(X, \|\cdot\|)$ satisfies

$$\bar{\beta}_{\|\cdot\|}(t_0) > 0 \quad \text{for some } t_0 \in (0, 1),$$

then X admits an equivalent norm with Rolewicz's property (β) . In the final paragraph, the authors observe that the asymptotic beta modulus also makes sense on the larger interval $[0, K(X))$, where

$$K(X) = \sup\{\lambda > 0 : \text{there is a sequence } (x_n) \subset B_X \text{ with } \inf_{n \neq m} \|x_n - x_m\| \geq \lambda\},$$

and ask:

It would be interesting to see if the condition in Theorem 1.2 could be pushed beyond 1.

for $\varepsilon \in (0, t_0)$ small enough, $\rho_{\|\cdot\|_\varepsilon}(t_0) > 0$, and yet $\rho_{\|\cdot\|_\varepsilon}(\varepsilon) = 0$.

The natural domain for the modulus of uniform convexity is clearly $[0, 2]$. Since every infinite-dimensional Banach space has a 1-separated sequence in its unit sphere it is natural to consider the interval $[0, 1]$ as domain of the modulus of property (β) . However, it also makes sense to consider this modulus on the interval $[0, K(X))$ where $K(X)$ is the Kottman's constant of X , i.e.,

$$(11) \quad K(X) := \sup\{\lambda > 0 : \exists (x_n)_{n \in \mathbb{N}} \subset B_X \text{ s.t. } \inf_{n \neq m} \|x_n - x_m\| \geq \lambda\}.$$

The non-trivial fact that $K(X) > 1$ is due to Elton and Odell [8]. It would be interesting to see if the condition in Theorem 1.2 could be pushed beyond 1.

Answer

The answer to the natural strengthened statement is negative. There is a non-reflexive Banach space whose given norm has positive asymptotic beta modulus at every scale strictly larger than 1 up to its Kottman range. Consequently, one-point positivity at some $t_0 \in (1, K(X))$ does not force the space to admit an equivalent property- (β) norm.

Definitions

For a Banach space $(X, \|\cdot\|)$ and a parameter for which separated sequences exist in the unit ball, use the extended asymptotic beta modulus

$$\bar{\beta}_X(t) = \inf \left\{ \sup_{n \in \mathbb{N}} \left(1 - \left\| \frac{x + y_n}{2} \right\| \right) : x \in B_X, (y_n) \subset B_X, \inf_{n \neq m} \|y_n - y_m\| \geq t \right\}.$$

Rolewicz's property (β) is equivalent to positivity of this modulus on $(0, 1]$ for the given norm. It is standard, and also recalled in the source paper, that property (β) implies reflexivity. Hence a non-reflexive space cannot admit any equivalent norm with property (β) .

Proof Intuition

Let

$$X = \ell_2 \oplus_2 c_0, \quad \|(h, z)\| = (\|h\|_2^2 + \|z\|_\infty^2)^{1/2}.$$

The c_0 summand makes X non-reflexive. However, c_0 by itself cannot support infinite separated sequences at any scale above 1: in any bounded sequence in c_0 , two terms are within $1 + \varepsilon$ in sup norm. Thus any sequence in X separated by a scale $t > 1$ must, for a suitable pair of terms, use a definite amount of separation in the Hilbert coordinate. The Hilbert coordinate is strictly convex enough that two such Hilbert-separated terms cannot both have almost norm-one midpoint with the same point $x \in B_X$.

Auxiliary Lemmas

Lemma 1 (The c_0 Kottman threshold). *Let (z_n) be a sequence in B_{c_0} . For every $c > 1$ there exist distinct indices n, m such that*

$$\|z_n - z_m\|_\infty \leq c.$$

Proof. Write $c = 1 + \varepsilon$ with $\varepsilon > 0$. Since $c_0 \subset [-1, 1]^\mathbb{N}$, a diagonal argument gives a subsequence, still denoted (z_n) , which converges coordinatewise. Choose a finite set $F \subset \mathbb{N}$ such that $|z_1(k)| \leq \varepsilon/2$ for $k \notin F$. By coordinatewise convergence, pick $m > 1$ so large that $|z_m(k) - z_1(k)| \leq \varepsilon$ for all $k \in F$. If $k \notin F$, then

$$|z_m(k) - z_1(k)| \leq |z_m(k)| + |z_1(k)| \leq 1 + \varepsilon/2 < 1 + \varepsilon.$$

Thus $\|z_m - z_1\|_\infty \leq 1 + \varepsilon = c$. □

Lemma 2 (Two-point Hilbert-sum midpoint drop). *Let H be a Hilbert space, let Z be a Banach space, and put $E = H \oplus_2 Z$. If $0 \leq c < t \leq 2$, then there is a number $\eta = \eta(c, t) > 0$ such that the following holds. Whenever*

$$x = (u, v), \quad y = (h, z), \quad y' = (h', z') \in B_E$$

satisfy

$$\|z - z'\|_Z \leq c \quad \text{and} \quad \|y - y'\|_E \geq t,$$

then

$$\max \left\{ 1 - \left\| \frac{x + y}{2} \right\|_E, 1 - \left\| \frac{x + y'}{2} \right\|_E \right\} \geq \eta.$$

Proof. Assume the contrary. Then there are triples $x_j = (u_j, v_j)$, $y_j = (h_j, z_j)$ and $y'_j = (h'_j, z'_j)$ in B_E such that

$$\|z_j - z'_j\|_Z \leq c, \quad \|y_j - y'_j\|_E \geq t,$$

while

$$\left\| \frac{x_j + y_j}{2} \right\|_E \rightarrow 1 \quad \text{and} \quad \left\| \frac{x_j + y'_j}{2} \right\|_E \rightarrow 1.$$

Equivalently, $\|x_j + y_j\|_E \rightarrow 2$ and $\|x_j + y'_j\|_E \rightarrow 2$.

From $\|x_j + y_j\|_E \leq \|x_j\|_E + \|y_j\|_E \leq 2$, we get $\|x_j\|_E \rightarrow 1$ and $\|y_j\|_E \rightarrow 1$; similarly $\|y'_j\|_E \rightarrow 1$. Put

$$A_j = (\|u_j\|_H, \|v_j\|_Z), \quad B_j = (\|h_j\|_H, \|z_j\|_Z), \quad C_j = (\|h'_j\|_H, \|z'_j\|_Z)$$

in the Euclidean plane. Since

$$\|x_j + y_j\|_E \leq \|A_j + B_j\|_2 \leq \|A_j\|_2 + \|B_j\|_2 \leq 2$$

and the left side tends to 2, equality is asymptotically attained in the Euclidean triangle inequality. Hence $\|A_j - B_j\|_2 \rightarrow 0$. The same argument with y'_j gives $\|A_j - C_j\|_2 \rightarrow 0$.

Moreover,

$$0 \leq (\|u_j\|_H + \|h_j\|_H)^2 - \|u_j + h_j\|_H^2 \leq \|A_j + B_j\|_2^2 - \|x_j + y_j\|_E^2 \rightarrow 0.$$

Using the Hilbert identity

$$\|u_j - h_j\|_H^2 = (\|u_j\|_H - \|h_j\|_H)^2 + ((\|u_j\|_H + \|h_j\|_H)^2 - \|u_j + h_j\|_H^2),$$

we obtain $\|u_j - h_j\|_H \rightarrow 0$. Repeating the same argument for y'_j gives $\|u_j - h'_j\|_H \rightarrow 0$, and therefore $\|h_j - h'_j\|_H \rightarrow 0$.

It follows that

$$\limsup_j \|y_j - y'_j\|_E^2 = \limsup_j \left(\|h_j - h'_j\|_H^2 + \|z_j - z'_j\|_Z^2 \right) \leq c^2 < t^2,$$

contradicting $\|y_j - y'_j\|_E \geq t$ for all j . □

Main Counterexample

Theorem 1. *Let $X = \ell_2 \oplus_2 c_0$ with its natural ℓ_2 -sum norm. Then:*

1. $K(X) \geq \sqrt{2}$.
2. $\bar{\beta}_X(t) > 0$ for every $t \in (1, \sqrt{2})$.
3. X does not admit an equivalent norm with property (β) .

In particular, X satisfies $\bar{\beta}_X(t_0) > 0$ for some $t_0 \in (1, K(X))$ but is not property- (β) renormable.

Proof. For (1), let (e_n) be an orthonormal sequence in ℓ_2 . The points $(e_n, 0)$ lie in B_X and satisfy

$$\|(e_n, 0) - (e_m, 0)\|_X = \sqrt{2} \quad (n \neq m).$$

Thus $K(X) \geq \sqrt{2}$.

For (2), fix $t \in (1, \sqrt{2})$ and choose c with $1 < c < t$, for instance $c = (1+t)/2$. Let $\eta = \eta(c, t) > 0$ be given by Lemma 2. We prove that $\bar{\beta}_X(t) \geq \eta$.

Take an arbitrary $x = (u, v) \in B_X$ and an arbitrary sequence $y_n = (h_n, z_n) \in B_X$ with

$$\inf_{n \neq m} \|y_n - y_m\|_X \geq t.$$

By Lemma 1, there are distinct indices n, m such that

$$\|z_n - z_m\|_\infty \leq c.$$

Since also $\|y_n - y_m\|_X \geq t$, Lemma 2 applied in $H = \ell_2$ and $Z = c_0$ yields

$$\max \left\{ 1 - \left\| \frac{x + y_n}{2} \right\|_X, 1 - \left\| \frac{x + y_m}{2} \right\|_X \right\} \geq \eta.$$

Consequently

$$\sup_k \left(1 - \left\| \frac{x + y_k}{2} \right\|_X \right) \geq \eta.$$

Because x and the separated sequence (y_n) were arbitrary, the defining infimum gives $\bar{\beta}_X(t) \geq \eta > 0$.

For (3), X contains the closed subspace $\{0\} \oplus c_0$, so X is non-reflexive. If X admitted an equivalent norm with property (β) , then X would be reflexive, since property (β) implies reflexivity. This contradiction proves that no equivalent property- (β) norm exists. $\square$

Verification Notes

The proof is deterministic and uses no numerical computation. The two points where human review should focus are:

- Lemma 1, which records exactly that c_0 has no infinite unit-ball separated sequence at a scale strictly larger than 1.
- Lemma 2, where near norm-one midpoints in $H \oplus_2 Z$ force the Hilbert parts of both points to converge to the Hilbert part of the common base point.

Together these lemmas give a uniform midpoint drop for every $t > 1$ and hence positive asymptotic beta modulus at any chosen $t_0 \in (1, \sqrt{2})$.

Novelty and Search Bounds

Before promotion, the run indexes were searched for arXiv:2401.17465, for the phrases **Kottman**, **pushed beyond 1**, and **property beta**, and for the earlier local attempt on the Kottman-domain beta extension. No durable solution packet or proof-gap packet for this target was found; the previous failed attempt is copied into this packet under **history/**.

Bounded web searches on June 26, 2026 for "2401.17465" "pushed beyond 1", "property beta" "Kottman's constant", "Rolewicz" "Kottman" "beta" "c0", and the exact paper title returned the source arXiv page and no explicit later answer to the Kottman-domain question. Novelty confidence is therefore moderate: the construction is short and should be checked against the specialist renorming literature, but no known answer was found in the bounded search.

Human Review Recommendation

Treat this as a likely valid counterexample to the strengthened Kottman-domain version of Theorem 1.2. A reviewer should verify that the source question is intended to include the natural formulation “some $t_0 \in (1, K(X))$ ” and should audit Lemma 2’s compactness argument. If both checks pass, the example $X = \ell_2 \oplus_2 c_0$ gives a negative answer.

References

- [1] F. P. Baudier and G. Lancien, *An asymptotic analog of a local-to-global phenomenon for uniformly convex renormings*, arXiv:2401.17465.
- [2] S. Rolewicz, *On drop property*, Studia Math. 85 (1987), 27–35.
- [3] D. Kutzarova, *k - β and k -nearly uniformly convex Banach spaces*, J. Math. Anal. Appl. 162 (1991), no. 2, 322–338.
- [4] J. Elton and E. Odell, *The unit ball of every infinite-dimensional normed linear space contains a $(1 + \epsilon)$ -separated sequence*, Colloq. Math. 44 (1981), 105–109.